

TSUIKI AB, Japan

WMO: 478400

Lat: 33.685N

Lon: 131.040E

Elev: 17

StdP: 101.12

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -1.0	(d) -8.0	(e) 1.9	(f) 3.5	(g) -6.8	(h) 2.1	(i) 3.6	(j) 11.4	(k) 4.9	(l) 10.3	(m) 5.8	(n) 3.5	(o) 240	(p) 0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.2	(c) 32.9	(d) 25.7	(e) 31.8	(f) 25.6	(g) 30.8	(h) 25.6	(i) 26.7	(j) 30.4	(k) 26.4	(l) 30.0	(m) 25.9	(n) 29.5	(o) 5.1	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	29.2	25.1	20.3	28.6	24.9	19.9	28.4	84.4	30.8	82.2	30.0	80.0	29.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.1	8.1	-4.0	34.5	0.9	1.1	-4.7	35.3	-5.2	35.9	-5.7	36.5	-6.4	37.3
			-5.1	27.6	0.8	0.6	-5.6	28.1	-6.1	28.4	-6.5	28.8	-7.1	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	5.0	5.6	8.7	13.4	18.2	22.1	26.3	27.2	23.5	18.0	12.3	7.2
	DBStd	8.14	2.41	2.68	2.98	3.03	2.35	2.21	2.15	1.69	2.49	2.74	3.15	2.84
	HDD10.0	460	157	126	62	7	0	0	0	0	0	0	13	95
	HDD18.3	1819	414	357	300	150	32	1	0	0	0	39	183	344
	CDD10.0	2531	1	3	21	108	253	364	506	534	405	247	81	8
	CDD18.3	849	0	0	0	2	27	114	247	276	155	28	1	0
	CDH23.3	6980	0	0	1	13	109	506	2309	2893	1055	93	2	0
CDH26.7	2147	0	0	0	2	8	75	766	1069	223	4	0	0	
Wind	WSAvg	3.5	3.8	3.6	3.8	3.5	3.4	3.2	3.1	3.5	3.5	3.4	3.4	3.7
Precipitation	PrecAvg	1951	62	81	121	144	168	369	318	198	223	119	89	60
	PrecMax	3102	181	157	340	256	305	690	692	389	508	361	196	126
	PrecMin	1168	18	27	46	41	55	71	71	53	49	14	23	4
	PrecStd	382	37	33	56	53	64	154	178	103	131	88	46	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	16.9	21.0	25.0	27.9	30.9	34.1	34.2	32.0	27.1	22.0	17.0
		MCWB	10.3	13.4	14.6	16.5	17.8	24.6	25.3	25.6	25.4	21.8	17.3	13.4
	2%	DB	12.2	13.9	17.9	22.1	25.9	28.8	32.8	33.0	30.1	25.8	20.2	15.1
		MCWB	8.5	9.7	12.0	14.8	17.5	23.0	25.5	25.8	24.9	20.9	16.3	11.4
	5%	DB	11.0	12.1	15.9	20.2	24.2	27.2	31.2	32.0	29.0	24.1	19.1	14.0
		MCWB	7.6	8.3	11.0	14.4	17.5	22.1	25.5	25.7	24.1	19.6	15.6	10.5
	10%	DB	9.9	10.9	14.0	19.0	23.1	26.1	30.2	31.1	27.9	23.1	18.0	12.8
		MCWB	6.9	7.6	10.2	13.8	17.3	21.6	25.3	25.8	23.3	18.6	14.7	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	13.7	15.5	19.2	21.3	25.6	27.4	27.7	26.7	23.7	19.5	14.7
		MCDB	12.8	16.1	18.8	21.7	23.7	28.7	30.5	30.9	29.9	25.8	21.0	16.1
	2%	WB	9.8	11.2	13.3	17.2	20.2	24.5	26.6	27.0	25.8	22.1	17.6	12.6
		MCDB	11.0	13.0	16.0	19.9	22.8	27.3	30.3	30.8	28.7	24.4	19.3	14.5
	5%	WB	8.3	9.7	12.0	16.0	19.4	23.6	26.2	26.6	25.1	20.7	16.5	11.4
		MCDB	10.0	11.0	14.8	18.7	22.2	26.1	29.9	30.3	27.8	23.2	18.3	13.2
	10%	WB	7.1	8.3	10.8	15.1	18.5	22.7	25.7	26.2	24.4	19.6	15.3	10.1
		MCDB	9.1	10.0	13.3	17.8	21.6	25.0	29.3	29.9	27.0	22.1	17.4	12.1
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.7	9.2	9.7	9.1	6.7	6.5	7.2	7.3	8.6	9.0	8.6
		MCDBR	9.9	10.6	11.8	11.8	11.2	8.4	8.3	8.4	8.0	9.2	9.5	10.1
		MCWBR	6.9	7.5	7.5	6.8	6.2	4.2	3.6	3.7	4.1	5.4	6.4	7.2
	5% WB	MCDBR	8.9	8.7	9.9	9.0	8.1	6.1	7.2	7.4	6.7	7.4	7.9	8.7
		MCWBR	7.3	7.4	7.4	6.1	5.4	3.6	3.7	3.8	4.0	5.0	6.2	7.0
Clear-Sky Solar Irradiance	taub	0.427	0.469	0.531	0.552	0.558	0.571	0.570	0.535	0.504	0.459	0.430	0.420	
	taud	2.065	1.991	1.843	1.846	1.893	1.944	1.991	2.083	2.126	2.159	2.151	2.103	
	Ebn at Noon	749	761	746	754	755	743	741	760	764	766	745	731	
	Edh at Noon	132	158	197	206	198	188	178	160	147	132	120	121	
All-Sky Solar Radiation	RadAvg	2.21	2.91	3.99	4.81	5.21	4.44	4.86	5.11	4.07	3.43	2.57	2.03	
	RadStd	0.20	0.28	0.27	0.42	0.59	0.53	0.72	0.55	0.43	0.36	0.23	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.33	N/A	N/A	N/A	-0.46	-0.69	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page